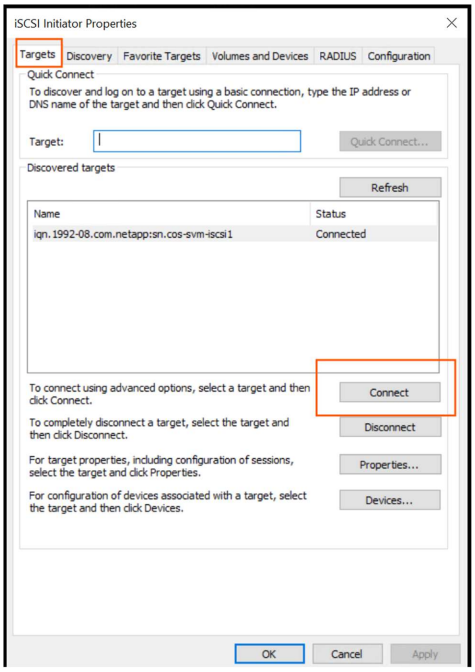
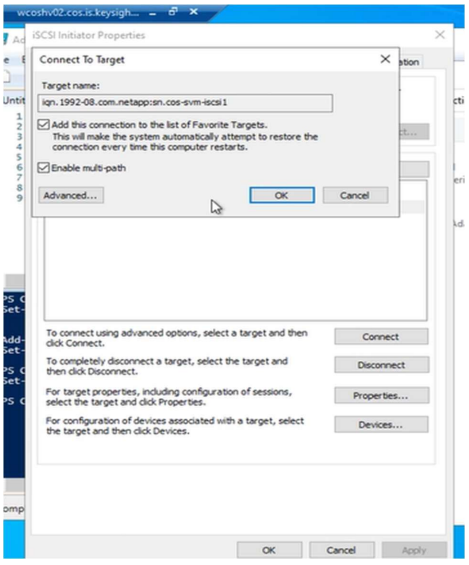
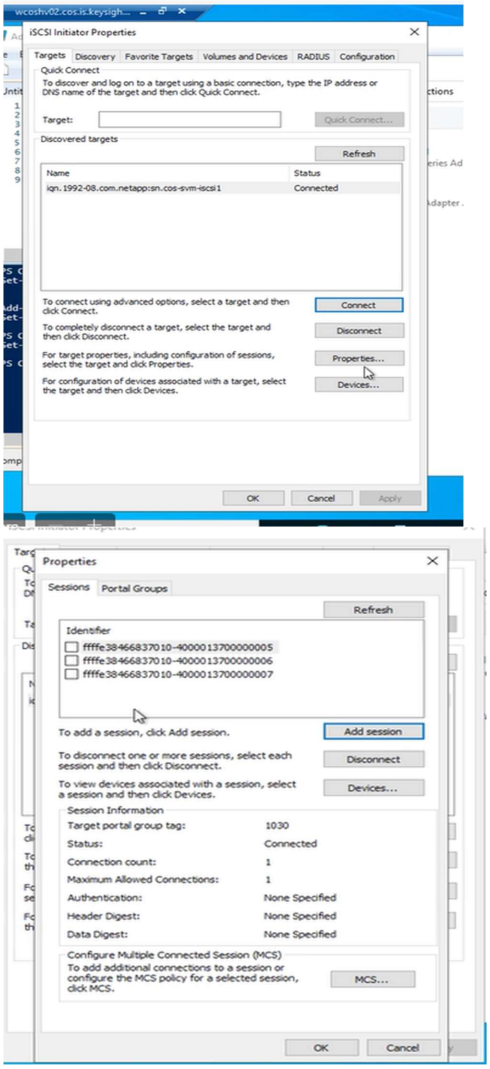
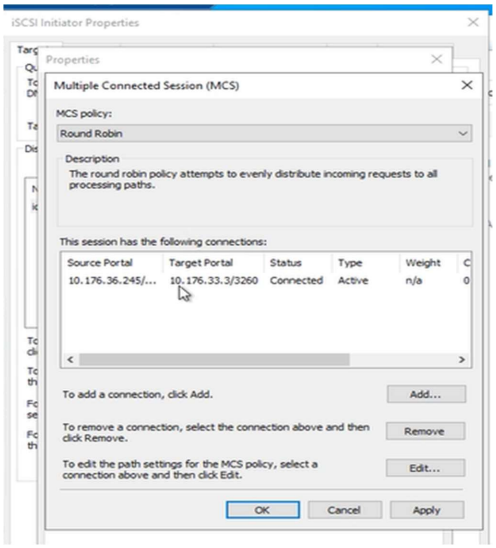

iSCSI Initiator Properties
Targets
Discovery
Favorite Targets
Volumes and Devices
RADIUS
Configuration
Quick Connect
To discover and log on to a target using a basic connection, type the IP address or DNS name of the target and then click Quick Connect.
Target:
Quick Connect...
Discovered targets
Refresh
Name
Status
iqn.1992-08.com.netapp:sn.cos-svm-iscsi1
Connected
To connect using advanced options, select a target and then click Connect.
Connect
To completely disconnect a target, select the target and then click Disconnect.
Disconnect
For target properties, including configuration of sessions, select the target and click Properties.
Properties...
For configuration of devices associated with a target, select the target and then click Devices.
Devices...
OK
Cancel
Apply


iSCSI Initiator Properties
Connect To Target
Target name:
iqn.1992-08.com.netapp:sn.cos-svm-iscsi1
Add this connection to the list of Favorite Targets.
This will make the system automatically attempt to restore the connection every time this computer restarts.
Enable multi-path
Advanced...
OK
Cancel
To connect using advanced options, select a target and then click Connect.
Connect
To completely disconnect a target, select the target and then click Disconnect.
Disconnect
For target properties, including configuration of sessions, select the target and click Properties.
Properties...
For configuration of devices associated with a target, select the target and then click Devices.
Devices...
OK
Cancel
Apply


wcoshv02.cos.is.keysigh...
iSCSI Initiator Properties
Targets
Discovery
Favorite Targets
Volumes and Devices
RADIUS
Configuration
Quick Connect
To discover and log on to a target using a basic connection, type the IP address or DNS name of the target and then click Quick Connect.
Target:
Quick Connect...
Discovered targets
Refresh
Name
Status
iqn.1992-08.com.netapp:sn.cos-svm-iscsi1
Connected
To connect using advanced options, select a target and then click Connect.
Connect
To completely disconnect a target, select the target and then click Disconnect.
Disconnect
For target properties, including configuration of sessions, select the target and click Properties.
Properties...
For configuration of devices associated with a target, select the target and then click Devices.
Devices...
OK
Cancel
Apply
Properties
Sessions
Portal Groups
Refresh
Identifier
ffffe3846683701 0-4000013700000005
ffffe3846683701 0-4000013700000006
ffffe3846683701 0-4000013700000007
To add a session, click Add session.
Add session
To disconnect one or more sessions, select each session and then click Disconnect.
Disconnect
To view devices associated with a session, select a session and then click Devices.
Devices...
Session Information
Target portal group tag:
1030
Status:
Connected
Connection count:
1
Maximum Allowed Connections:
1
Authentication:
None Specified
Header Digest:
None Specified
Data Digest:
None Specified
Configure Multiple Connected Session (MCS)
To add additional connections to a session or configure the MCS policy for a selected session, click MCS.
MCS...
OK
Cancel

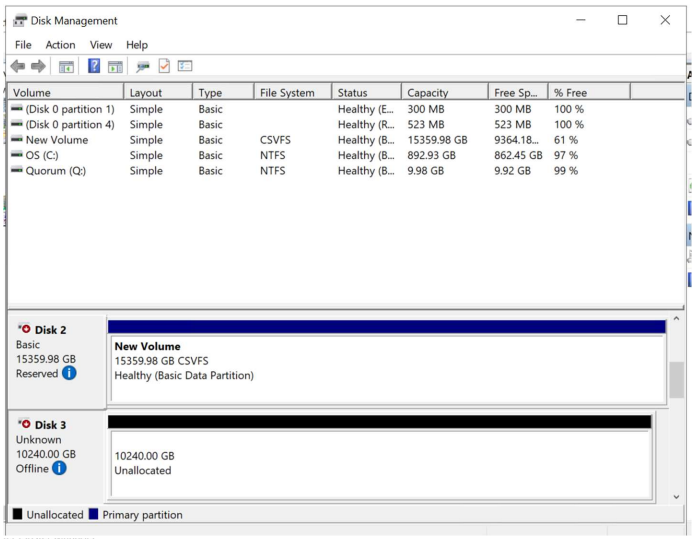


4.9 Configure Quorum Disk & Cluster Shared Volumes (CSV)

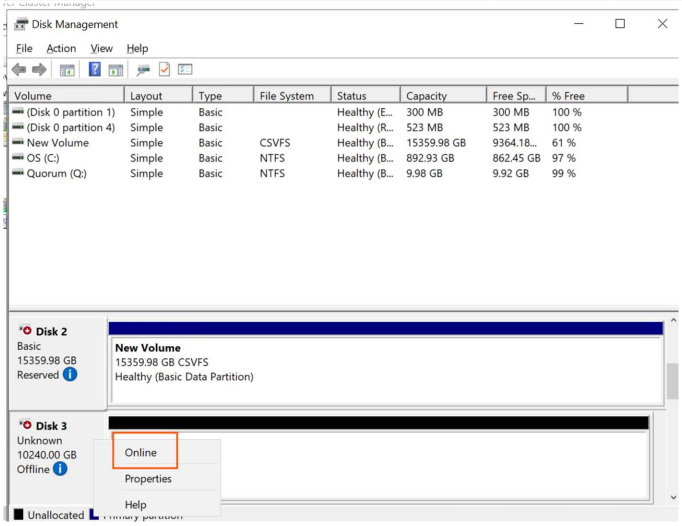
Quorum Disk Configuration:

1. Map LUN using ISCSI.
2. Enable added storage from windows disk manager.

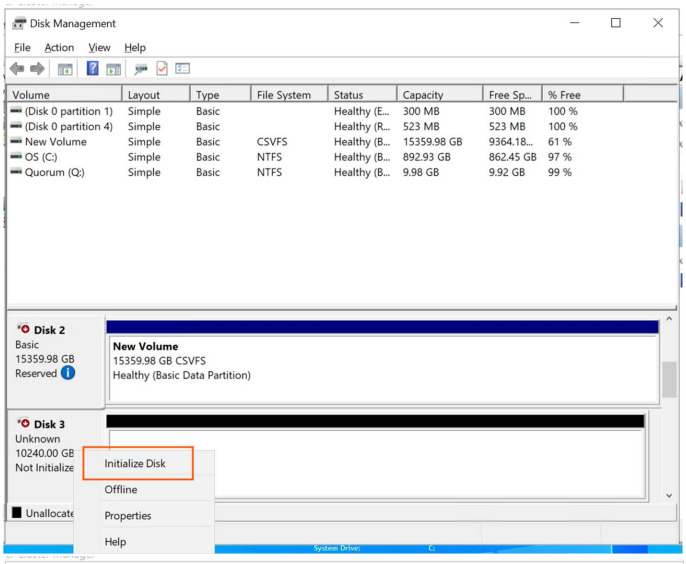
a.


Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
(Disk 0 partition 1) Simple Basic Healthy (E... 300 MB 300 MB 100 %
(Disk 0 partition 4) Simple Basic Healthy (R... 523 MB 523 MB 100 %
New Volume Simple Basic CSVFS Healthy (B... 15359.98 GB 9364.18... 61 %
OS (C:) Simple Basic NTFS Healthy (B... 892.93 GB 862.45 GB 97 %
Quorum (Q:) Simple Basic NTFS Healthy (B... 9.98 GB 9.92 GB 99 %
Disk 2
Basic
15359.98 GB
Reserved
New Volume
15359.98 GB CSVFS
Healthy (Basic Data Partition)
Disk 3
Unknown
10240.00 GB
Offline
10240.00 GB
Unallocated
Unallocated
Primary partition

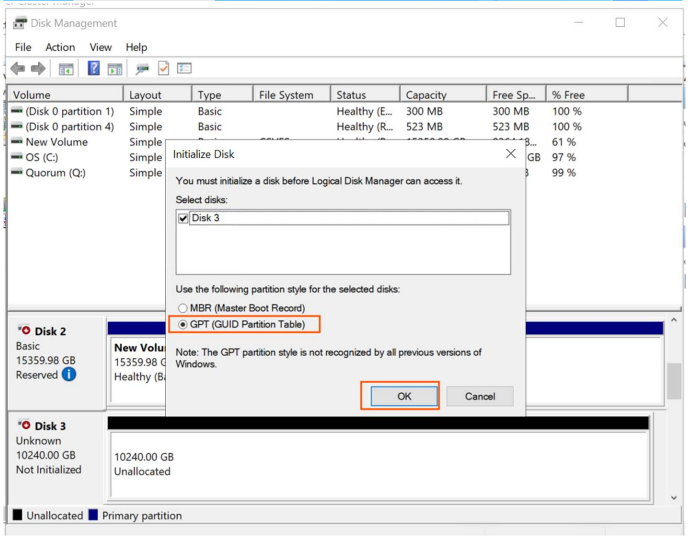
b.


Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
(Disk 0 partition 1) Simple Basic Healthy (E... 300 MB 300 MB 100 %
(Disk 0 partition 4) Simple Basic Healthy (R... 523 MB 523 MB 100 %
New Volume Simple Basic CSVFS Healthy (B... 15359.98 GB 9364.18... 61 %
OS (C:) Simple Basic NTFS Healthy (B... 892.93 GB 862.45 GB 97 %
Quorum (Q:) Simple Basic NTFS Healthy (B... 9.98 GB 9.92 GB 99 %
Disk 2
Basic
15359.98 GB
Reserved
New Volume
15359.98 GB CSVFS
Healthy (Basic Data Partition)
Disk 3
Unknown
10240.00 GB
Offline
Online
Properties
Help
Unallocated

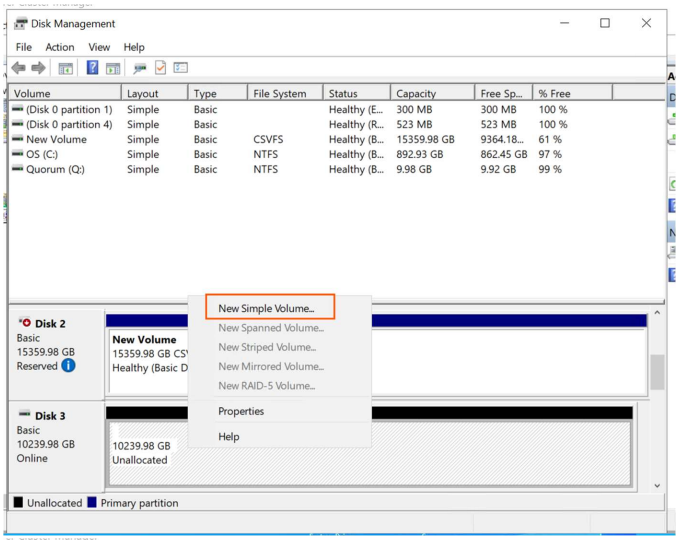
c.


Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
(Disk 0 partition 1) Simple Basic Healthy (E... 300 MB 300 MB 100 %
(Disk 0 partition 4) Simple Basic Healthy (R... 523 MB 523 MB 100 %
New Volume Simple Basic CSVFS Healthy (B... 15359.98 GB 9364.18... 61 %
OS (C:) Simple Basic NTFS Healthy (B... 892.93 GB 862.45 GB 97 %
Quorum (Q:) Simple Basic NTFS Healthy (B... 9.98 GB 9.92 GB 99 %
Disk 2
Basic
15359.98 GB
Reserved
New Volume
15359.98 GB CSVFS
Healthy (Basic Data Partition)
Disk 3
Unknown
10240.00 GB
Not Initialize
Initialize Disk
Offline
Properties
Help
Unallocate

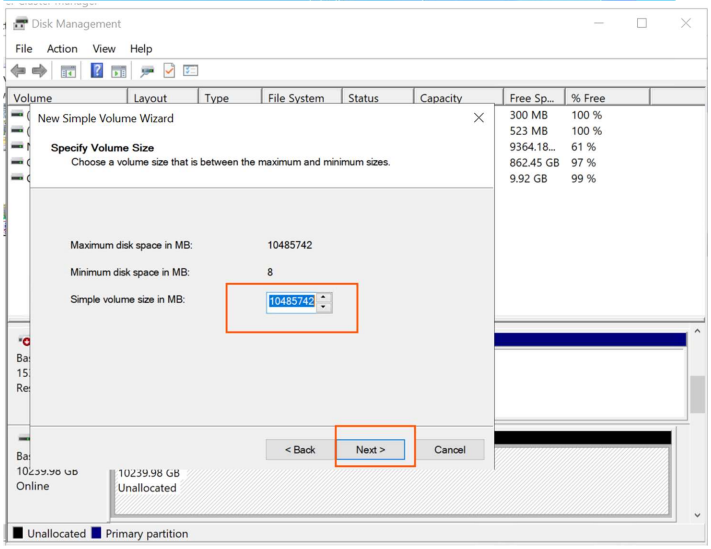
d.


Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
(Disk 0 partition 1) Simple Basic Healthy (E... 300 MB 300 MB 100 %
(Disk 0 partition 4) Simple Basic Healthy (R... 523 MB 523 MB 100 %
New Volume Simple 61 %
OS (C:) Simple 97 %
Quorum (Q:) Simple 99 %
Initialize Disk
You must initialize a disk before Logical Disk Manager can access it.
Select disks:
Disk 3
Use the following partition style for the selected disks:
MBR (Master Boot Record)
GPT (GUID Partition Table)
Note: The GPT partition style is not recognized by all previous versions of Windows.
OK
Cancel
Disk 2
Basic
15359.98 GB
Reserved
Disk 3
Unknown
10240.00 GB
Not Initialized
10240.00 GB
Unallocated
Unallocated
Primary partition

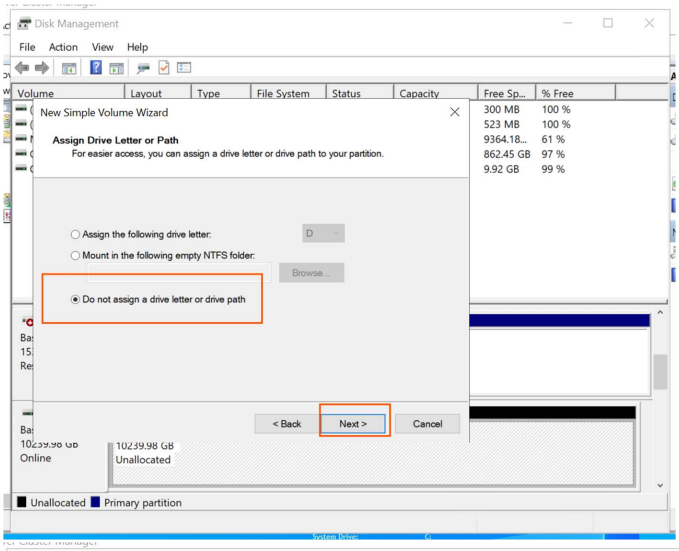
e.


Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
(Disk 0 partition 1) Simple Basic Healthy (E... 300 MB 300 MB 100 %
(Disk 0 partition 4) Simple Basic Healthy (R... 523 MB 523 MB 100 %
New Volume Simple Basic CSVFS Healthy (B... 15359.98 GB 9364.18... 61 %
OS (C:) Simple Basic NTFS Healthy (B... 892.93 GB 862.45 GB 97 %
Quorum (Q:) Simple Basic NTFS Healthy (B... 9.98 GB 9.92 GB 99 %
Disk 2
Basic
15359.98 GB
Reserved
New Volume
15359.98 GB
Healthy (Basic D
New Simple Volume...
New Spanned Volume...
New Striped Volume...
New Mirrored Volume...
New RAID-5 Volume...
Properties
Help
Disk 3
Basic
10239.98 GB
Online
10239.98 GB
Unallocated
Unallocated
Primary partition

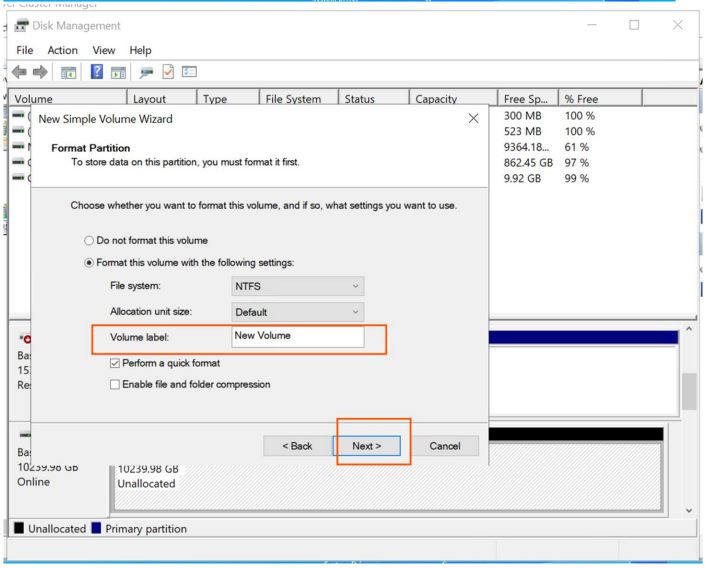
f.


Disk Management
File Action View Help
New Simple Volume Wizard
Specify Volume Size
Choose a volume size that is between the maximum and minimum sizes.
Maximum disk space in MB: 10485742
Minimum disk space in MB: 8
Simple volume size in MB: 10485742
< Back
Next >
Cancel
Online
10239.98 GB
Unallocated
Unallocated
Primary partition

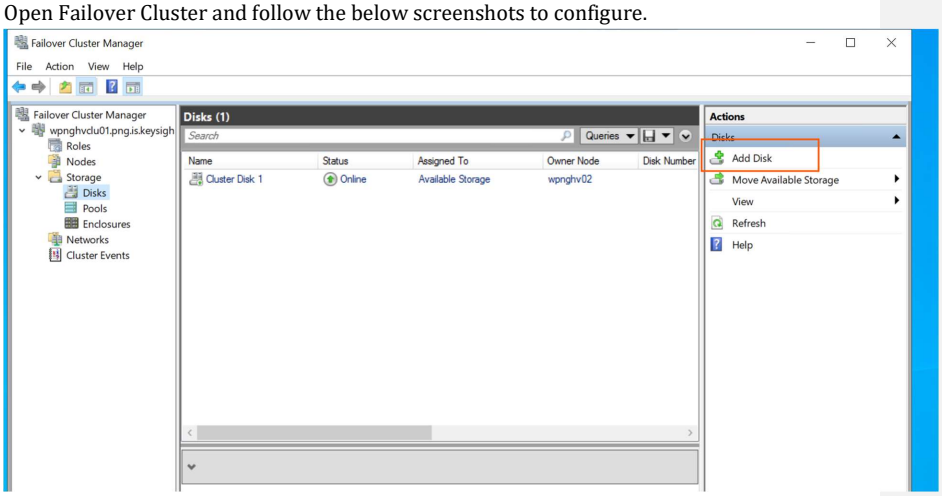
g.

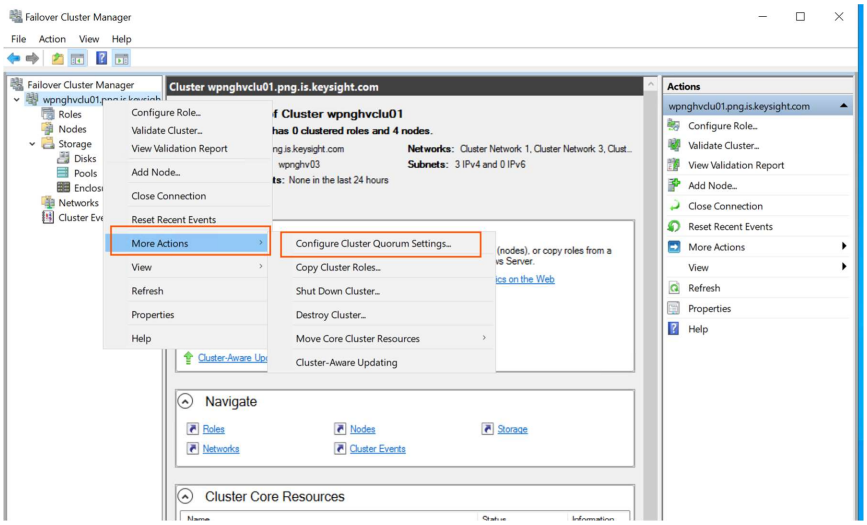

Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
300 MB 100 %
523 MB 100 %
9364.18... 61 %
862.45 GB 97 %
9.92 GB 99 %
New Simple Volume Wizard
Assign Drive Letter or Path
For easier access, you can assign a drive letter or drive path to your partition.
Assign the following drive letter: D
Mount in the following empty NTFS folder:
Browse...
Do not assign a drive letter or drive path
< Back Next > Cancel
Online
10239.98 GB
Unallocated
Unallocated Primary partition

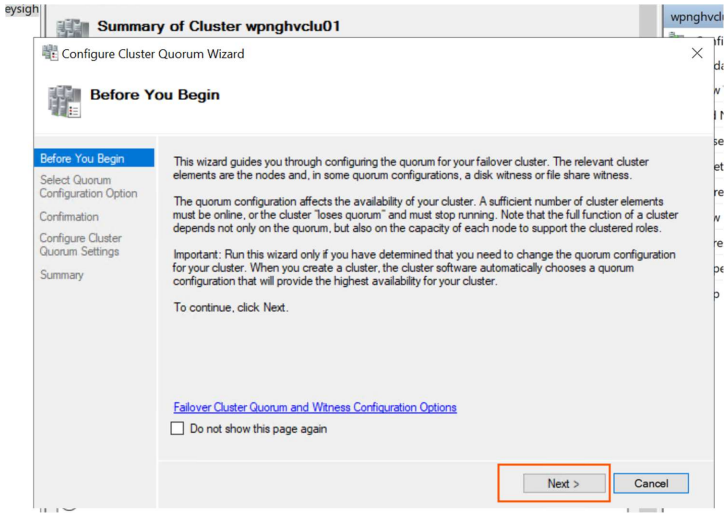
h.

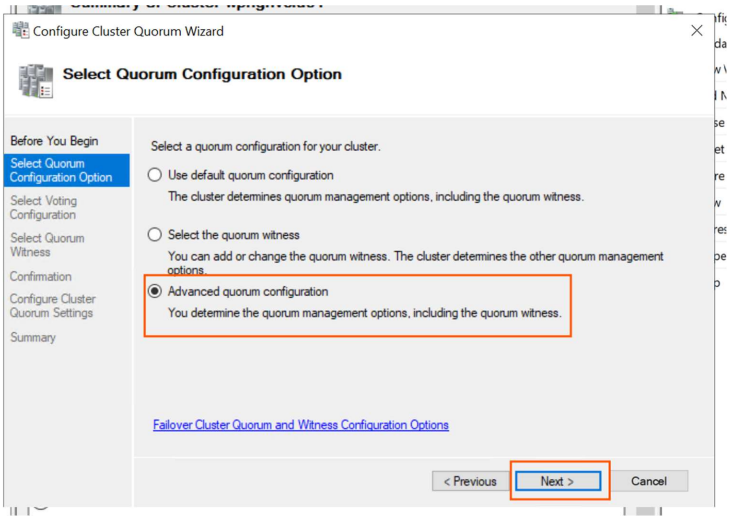

Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Sp... % Free
300 MB 100 %
523 MB 100 %
9364.18... 61 %
862.45 GB 97 %
9.92 GB 99 %
New Simple Volume Wizard
Format Partition
To store data on this partition, you must format it first.
Choose whether you want to format this volume, and if so, what settings you want to use.
Do not format this volume
Format this volume with the following settings:
File system: NTFS
Allocation unit size: Default
Volume label: New Volume
Perform a quick format
Enable file and folder compression
< Back Next > Cancel
Online
10239.98 GB
Unallocated
Unallocated Primary partition

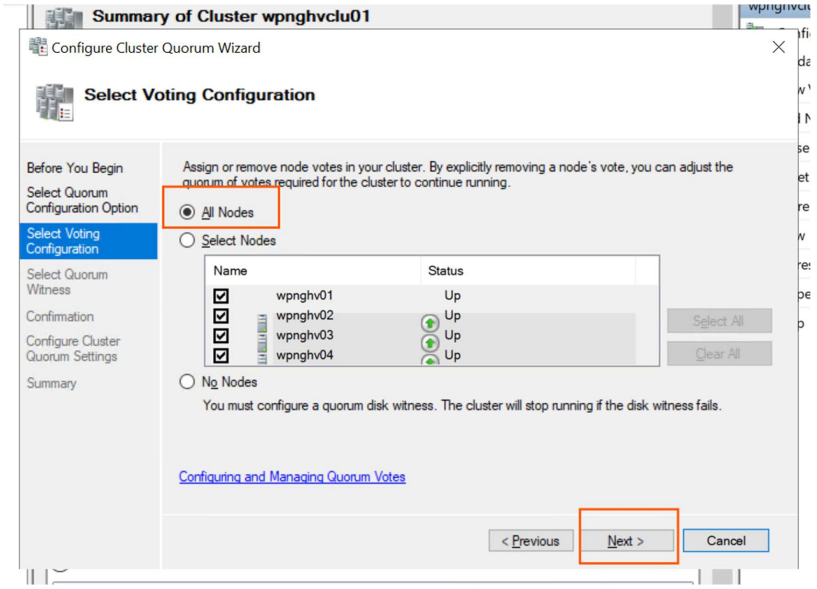
3. Open Failover Cluster and follow the below screenshots to configure.

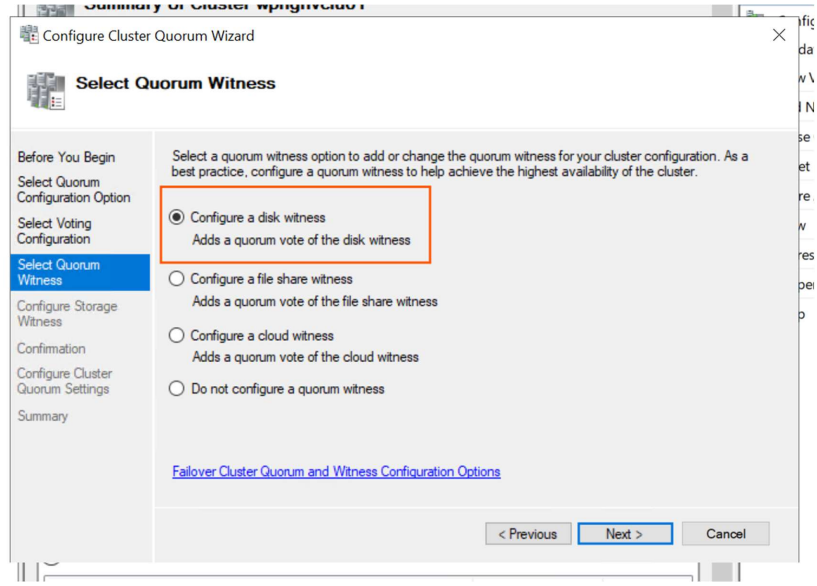


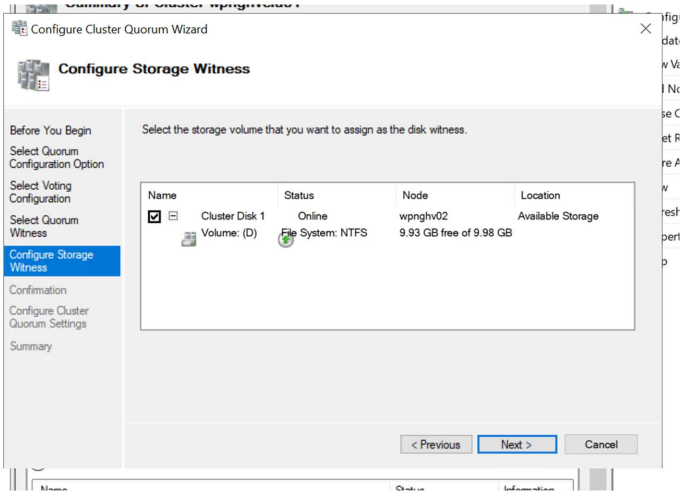


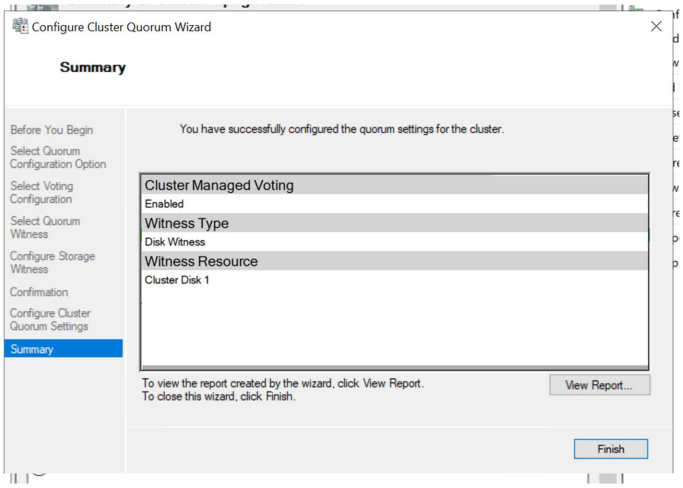

Summary of Cluster wpnghvclu01
Configure Cluster Quorum Wizard
Before You Begin
Before You Begin
Select Quorum Configuration Option
Confirmation
Configure Cluster Quorum Settings
Summary
This wizard guides you through configuring the quorum for your failover cluster. The relevant cluster elements are the nodes and, in some quorum configurations, a disk witness or file share witness.
The quorum configuration affects the availability of your cluster. A sufficient number of cluster elements must be online, or the cluster "loses quorum" and must stop running. Note that the full function of a cluster depends not only on the quorum, but also on the capacity of each node to support the clustered roles.
Important: Run this wizard only if you have determined that you need to change the quorum configuration for your cluster. When you create a cluster, the cluster software automatically chooses a quorum configuration that will provide the highest availability for your cluster.
To continue, click Next.
Failover Cluster Quorum and Witness Configuration Options
Do not show this page again
Next >
Cancel


Configure Cluster Quorum Wizard
Select Quorum Configuration Option
Before You Begin
Select Quorum Configuration Option
Select Voting Configuration
Select Quorum Witness
Confirmation
Configure Cluster Quorum Settings
Summary
Select a quorum configuration for your cluster.
Use default quorum configuration
The cluster determines quorum management options, including the quorum witness.
Select the quorum witness
You can add or change the quorum witness. The cluster determines the other quorum management options.
Advanced quorum configuration
You determine the quorum management options, including the quorum witness.
Failover Cluster Quorum and Witness Configuration Options
< Previous
Next >
Cancel


Summary of Cluster wpnghvclu01
Configure Cluster Quorum Wizard
Select Voting Configuration
Before You Begin
Select Quorum Configuration Option
Select Voting Configuration
Select Quorum Witness
Confirmation
Configure Cluster Quorum Settings
Summary
Assign or remove node votes in your cluster. By explicitly removing a node's vote, you can adjust the quorum of votes required for the cluster to continue running.
All Nodes
Select Nodes
Name
Status
wpnghv01
Up
wpnghv02
Up
wpnghv03
Up
wpnghv04
Up
Select All
Clear All
No Nodes
You must configure a quorum disk witness. The cluster will stop running if the disk witness fails.
Configuring and Managing Quorum Votes
< Previous
Next >
Cancel

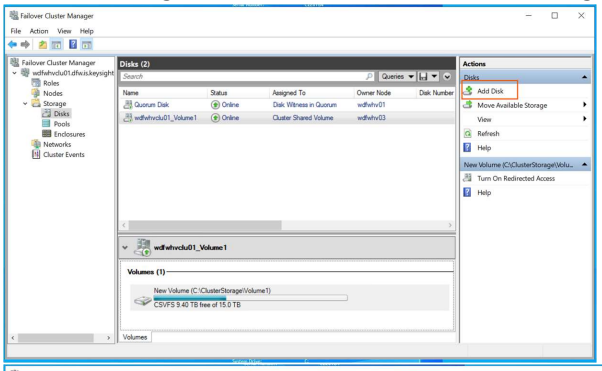

Configure Cluster Quorum Wizard
Select Quorum Witness
Before You Begin
Select Quorum Configuration Option
Select Voting Configuration
Select Quorum Witness
Configure Storage Witness
Confirmation
Configure Cluster Quorum Settings
Summary
Select a quorum witness option to add or change the quorum witness for your cluster configuration. As a best practice, configure a quorum witness to help achieve the highest availability of the cluster.
Configure a disk witness
Adds a quorum vote of the disk witness
Configure a file share witness
Adds a quorum vote of the file share witness
Configure a cloud witness
Adds a quorum vote of the cloud witness
Do not configure a quorum witness
Failover Cluster Quorum and Witness Configuration Options
< Previous
Next >
Cancel

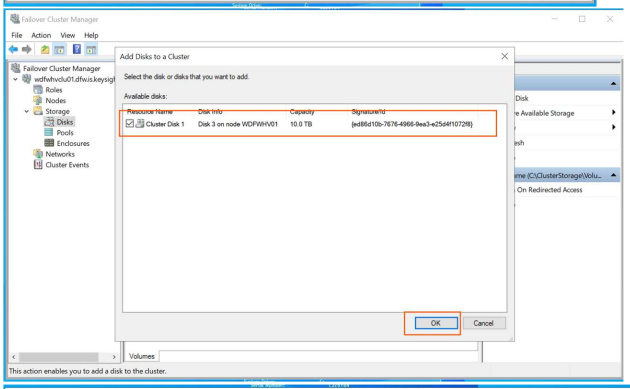

Configure Cluster Quorum Wizard
Configure Storage Witness
Before You Begin
Select Quorum Configuration Option
Select Voting Configuration
Select Quorum Witness
Configure Storage Witness
Confirmation
Configure Cluster Quorum Settings
Summary
Select the storage volume that you want to assign as the disk witness.
Name
Status
Node
Location
Cluster Disk 1
Online
wpnghv02
Available Storage
Volume: (D)
File System: NTFS
9.93 GB free of 9.98 GB
< Previous
Next >
Cancel

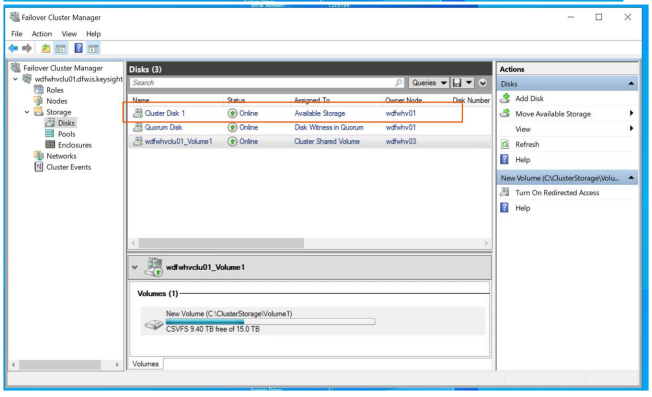

Configure Cluster Quorum Wizard
Summary
Before You Begin
Select Quorum Configuration Option
Select Voting Configuration
Select Quorum Witness
Configure Storage Witness
Confirmation
Configure Cluster Quorum Settings
Summary
You have successfully configured the quorum settings for the cluster.
Cluster Managed Voting
Enabled
Witness Type
Disk Witness
Witness Resource
Cluster Disk 1
To view the report created by the wizard, click View Report.
To close this wizard, click Finish.
View Report...
Finish

Cluster shared Volume (CSV) Configuration:

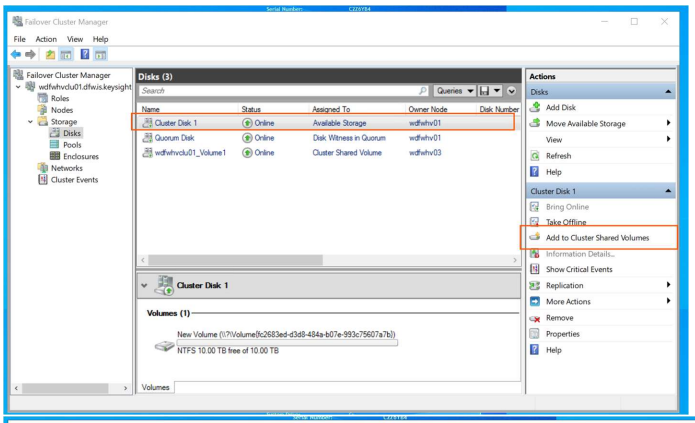
1. Connect the shared storage (SAN, iSCSI) to all cluster nodes. (Refer 4.8 step)
2. Enable added storage from windows disk manager. (Same as Quorum Disk Configuration Step 2)
3. In Failover Cluster Manager, select Add Disk and add the shared storage.

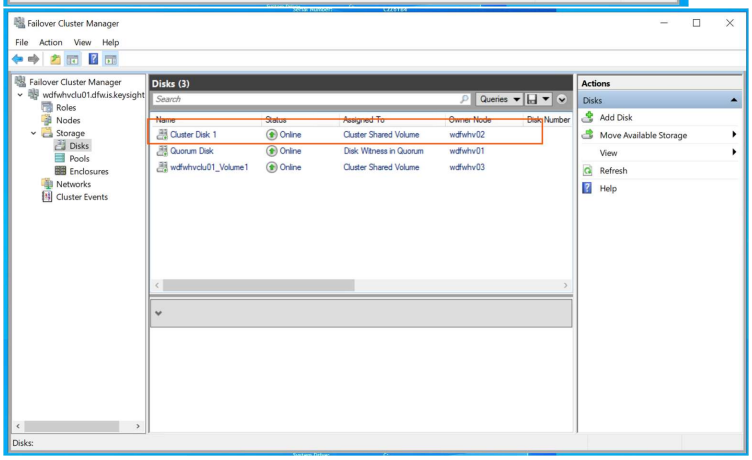






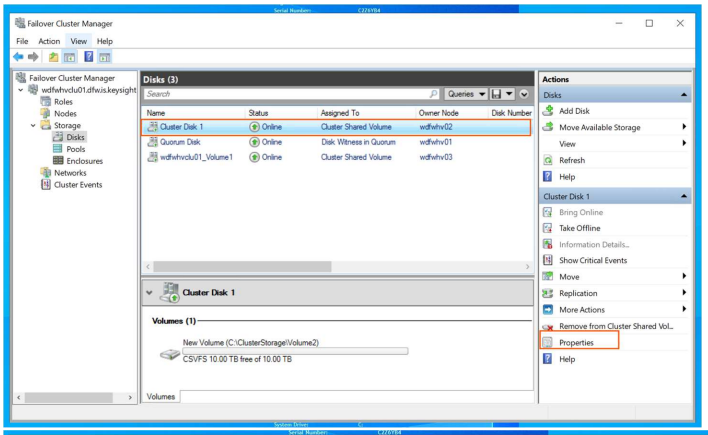
4. Right-click the disk and select Add to Cluster Shared Volumes.

a. 

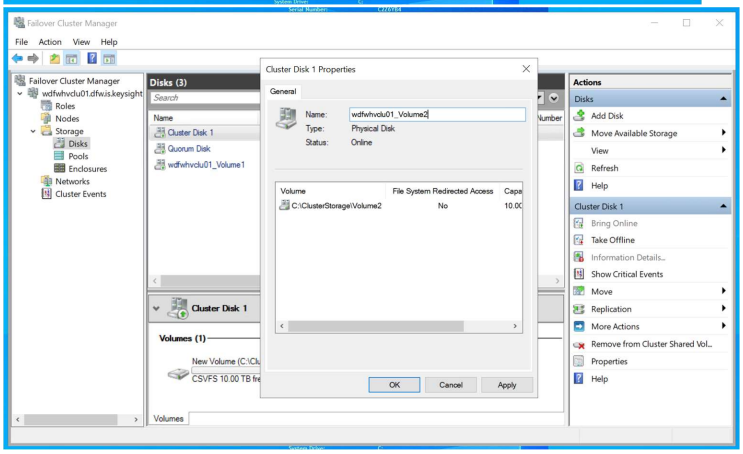
b. 

5. Rename the disk by clicking properties.
   Keysight Naming Standard : Clustername_Volume (ex: wcoshvclu02_volume2)

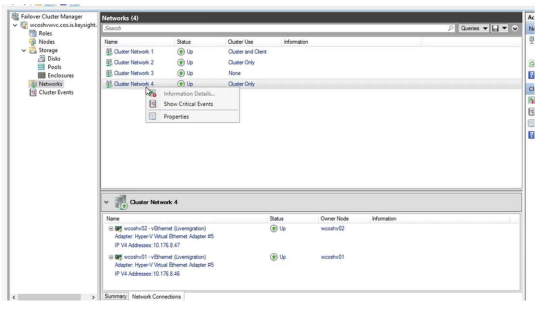
a.

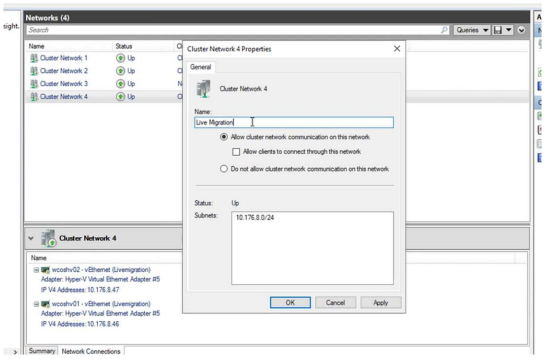


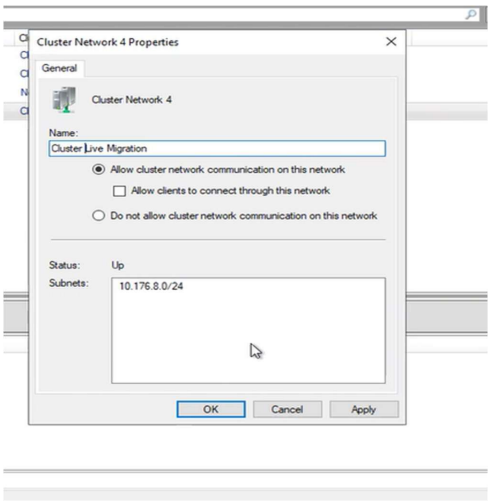
b.

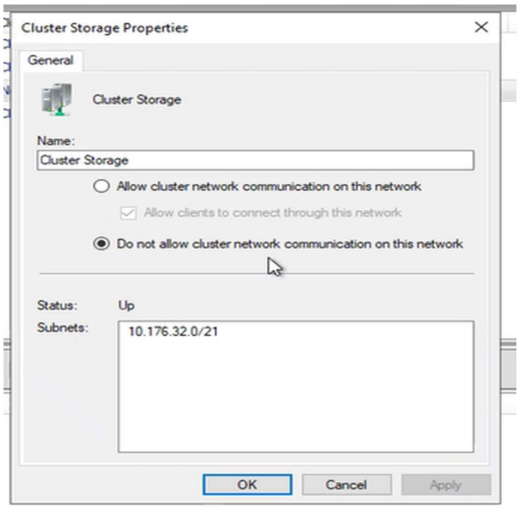


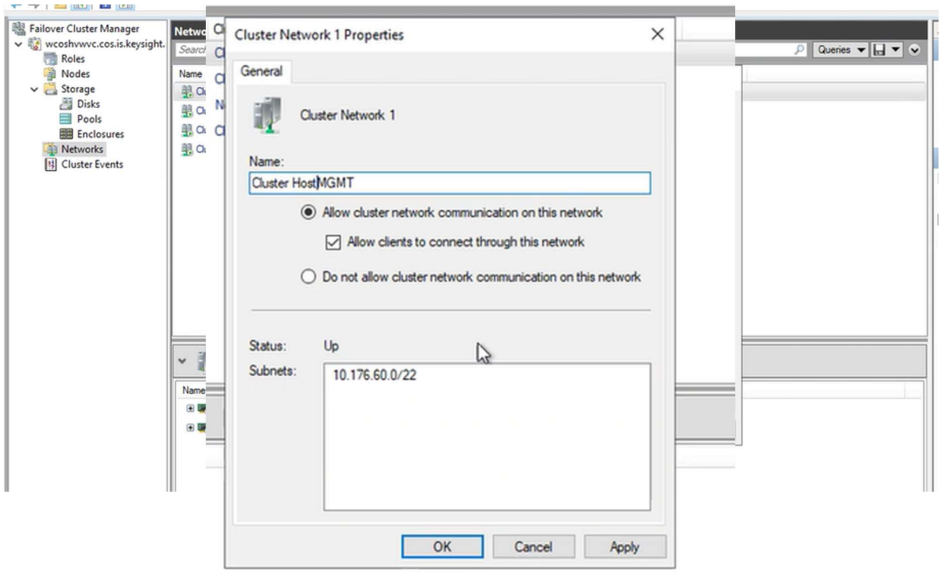
4.10. Naming Network Adapters




Networks (4)
Cluster Network 4 Properties
General
Cluster Network 4
Name:
Live Migration
Allow cluster network communication on this network
Allow clients to connect through this network
Do not allow cluster network communication on this network
Status: Up
Subnets: 10.176.8.0/24
OK
Cancel
Apply
Cluster Network 4
Summary
Network Connections


Cluster Network 4 Properties
General
Cluster Network 4
Name:
Cluster Live Migration
Allow cluster network communication on this network
Allow clients to connect through this network
Do not allow cluster network communication on this network
Status: Up
Subnets: 10.176.8.0/24
OK
Cancel
Apply


Cluster Storage Properties
General
Cluster Storage
Name:
Cluster Storage
Allow cluster network communication on this network
Allow clients to connect through this network
Do not allow cluster network communication on this network
Status:
Up
Subnets:
10.176.32.0/21
OK
Cancel
Apply

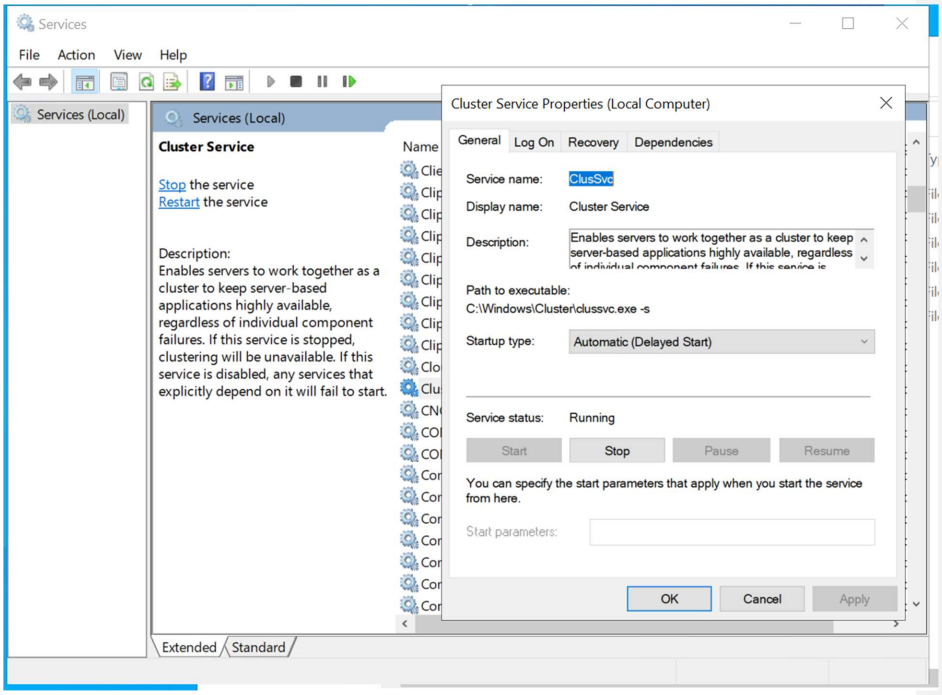


5. Post-Setup Configuration

5.1 Hyper-V Setting Configuration

➔ Set the Cluster Service to Automatic (Delayed Start) on all nodes in the cluster

- Right-Click on Start, select Run, type services.msc and click OK
- In Services double-click the Cluster Service, change the Startup type to Automatic (Delayed Start) and click OK.

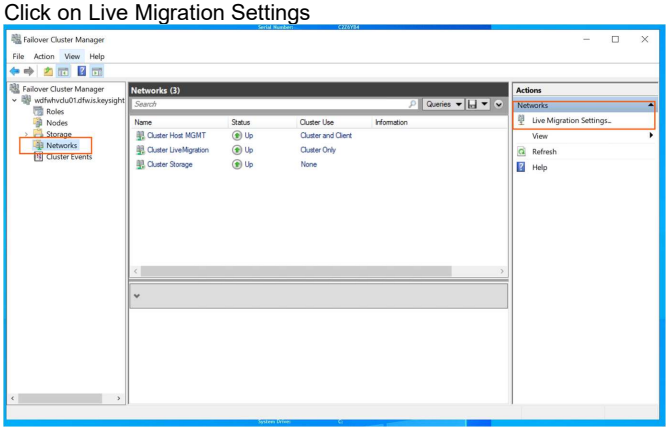
- 

➔ Remove host management network from live migration

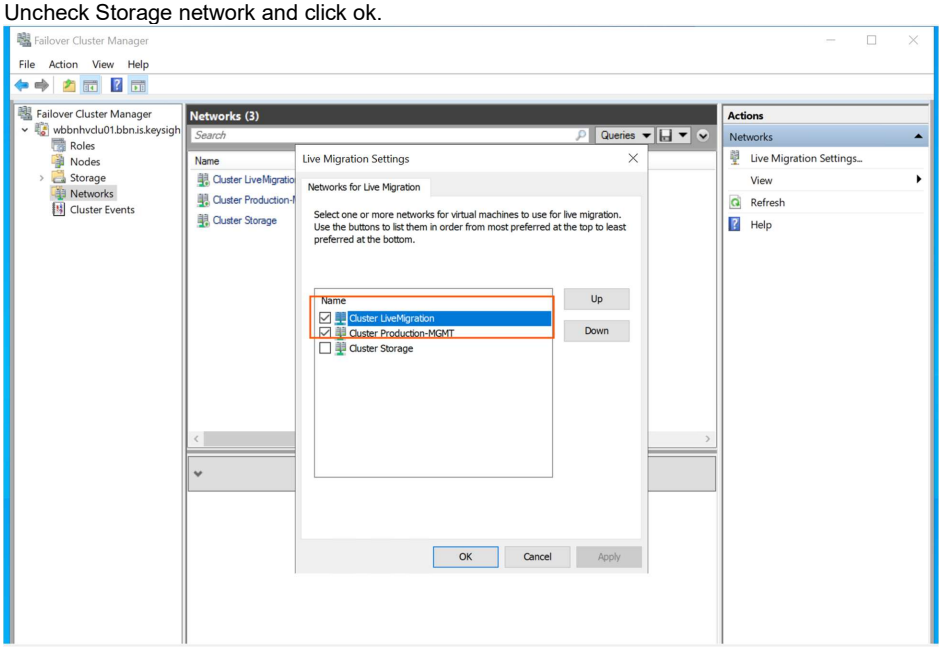
After you create the cluster, live migration is configured by default to use all available networks. Disable live migration on the host management network by excluding the host management network from the live migration settings.

Open Failover cluster and select Networks.

Click on Live Migration Settings

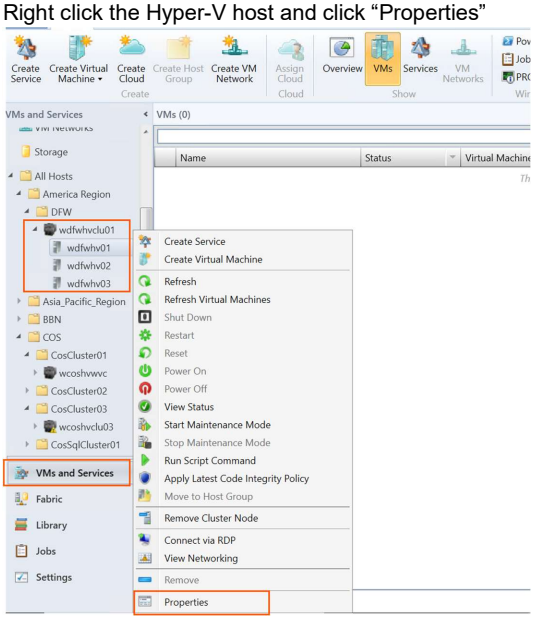


Uncheck Storage network and click ok.

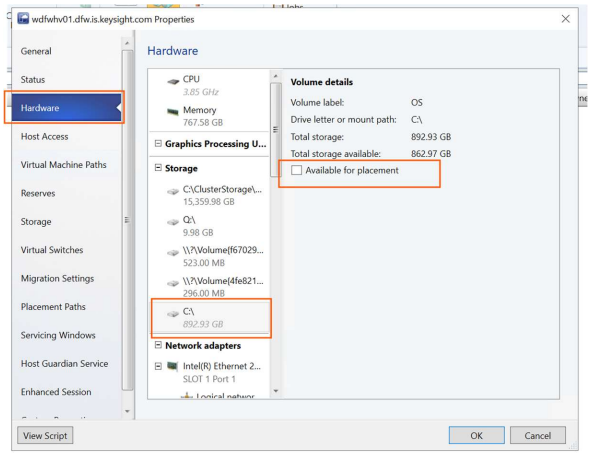


Configuring Disk settings

- In SCVMM open the “VMs and Services” workspace
- Right click the Hyper-V host and click “Properties”



- Select “Hardware”
- Under “Hardware” go to “Storage”
- Uncheck “Available for placement” on all not desired volumes, like all local volumes (C:\)

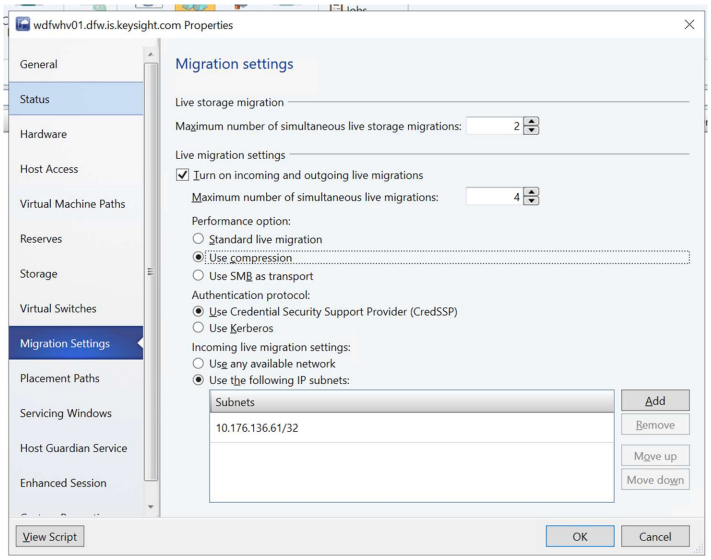


Configuring Cluster and Live Migration network settings

In SCVMM, on all nodes of the cluster, right-click → Properties → Migration settings

Configure the settings as shown:

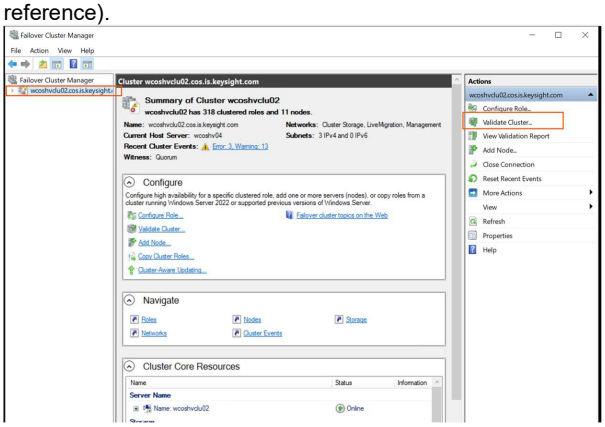
- Maximum live storage migrations: 2
- Incoming and outgoing live migrations selected
- Maximum simultaneous live migrations: 4
- Use Compression as Performance Options
- Use CredSSP as Authentication Protocol
- Use the following IP subnets selected

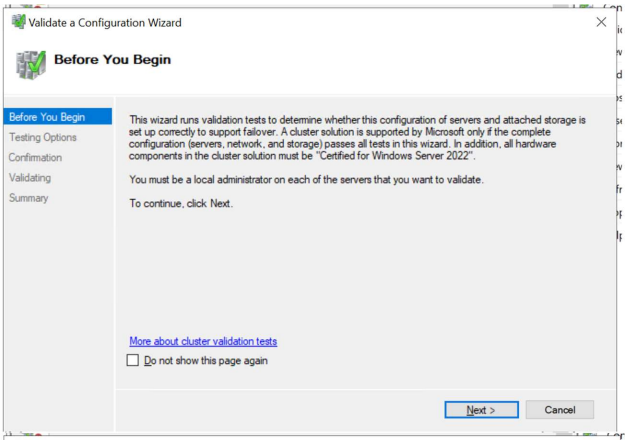
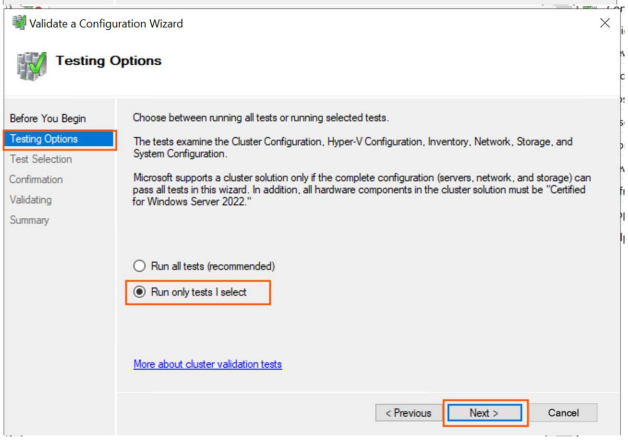
- 

6. Cluster Validation and recommendations

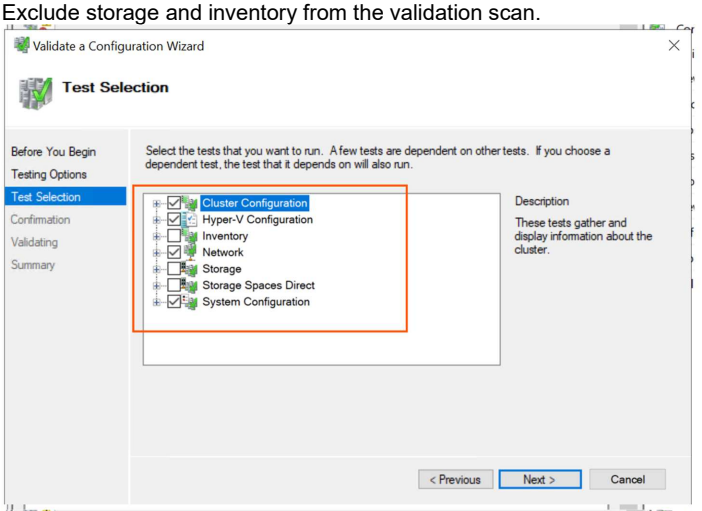
6.1 Cluster Validation report

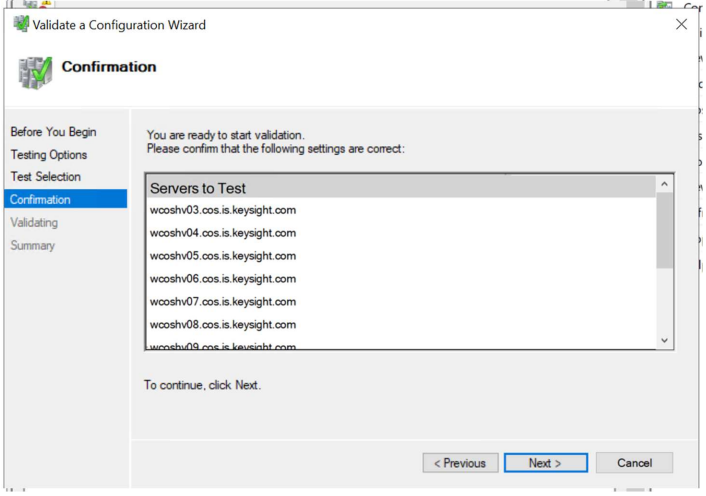
Once all setup completed, validate cluster by running cluster validation report.

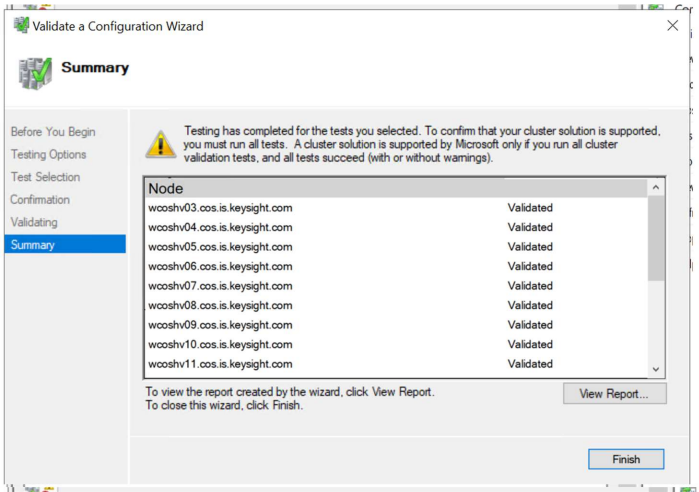
- Open failover cluster in any of the node.
- Select on cluster name from the left side menu.
- Click on Validate cluster option from the right side menu. (below screenshot for reference).
- 

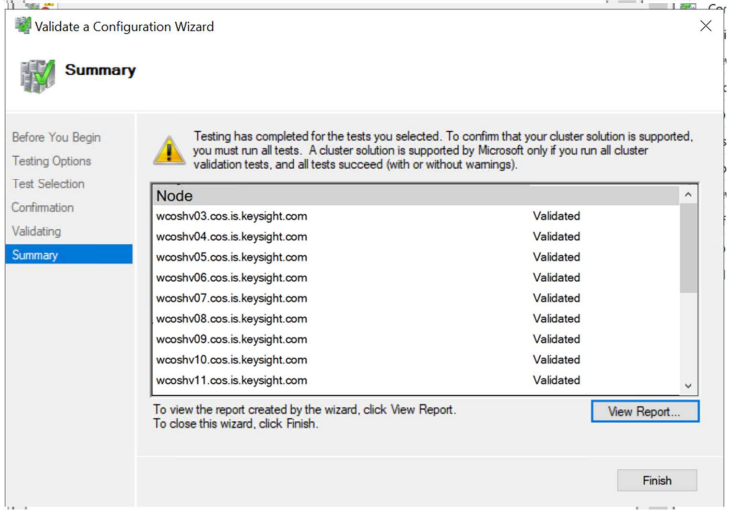
- 
Validate a Configuration Wizard
Before You Begin
Before You Begin
Testing Options
Confirmation
Validating
Summary
This wizard runs validation tests to determine whether this configuration of servers and attached storage is set up correctly to support failover. A cluster solution is supported by Microsoft only if the complete configuration (servers, network, and storage) passes all tests in this wizard. In addition, all hardware components in the cluster solution must be "Certified for Windows Server 2022".
You must be a local administrator on each of the servers that you want to validate.
To continue, click Next.
More about cluster validation tests
Do not show this page again
Next >
Cancel
- 
Validate a Configuration Wizard
Testing Options
Before You Begin
Testing Options
Test Selection
Confirmation
Validating
Summary
Choose between running all tests or running selected tests.
The tests examine the Cluster Configuration, Hyper-V Configuration, Inventory, Network, Storage, and System Configuration.
Microsoft supports a cluster solution only if the complete configuration (servers, network, and storage) can pass all tests in this wizard. In addition, all hardware components in the cluster solution must be "Certified for Windows Server 2022."
Run all tests (recommended)
Run only tests I select
More about cluster validation tests
< Previous
Next >
Cancel

- Exclude storage and inventory from the validation scan.





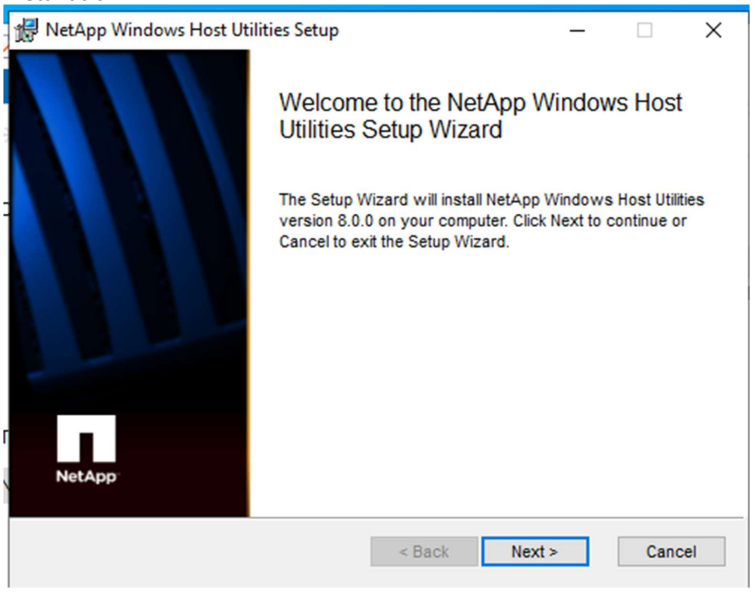
- 

- 

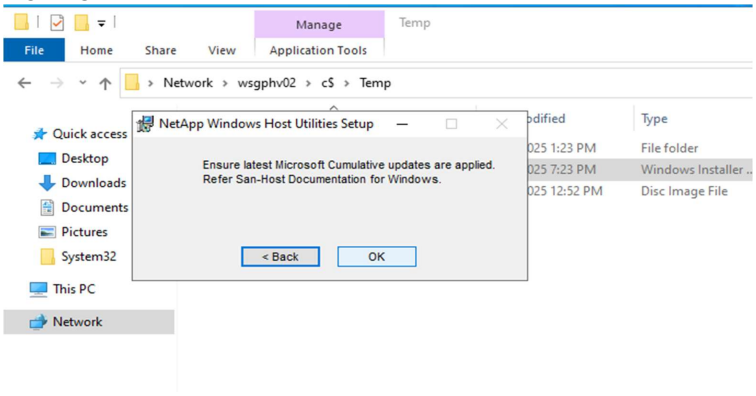
- Attach validation report to release the host in ASAP.

6.2 NetApp Windows Host Utilities Installation Process

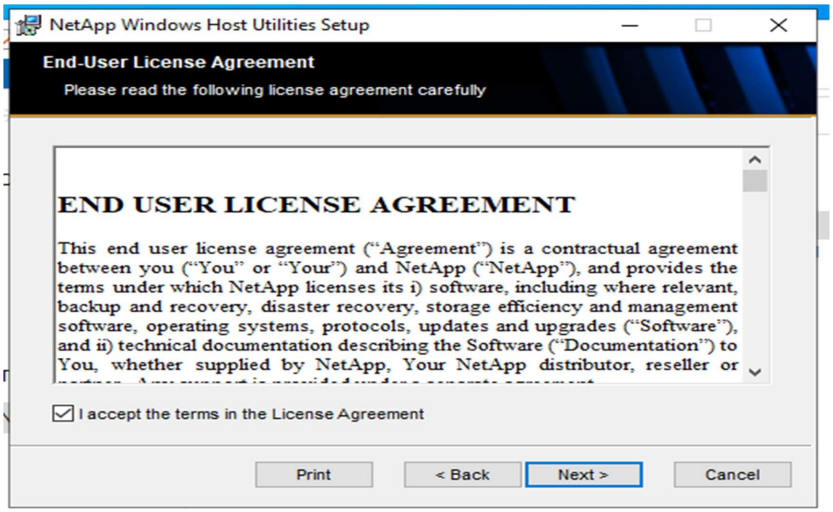
Application Location: \\wsgphv02.sgp.is.keysight.com\c$\Temp
Installation Process:
1.right click on application and run as administrator and click next to proceed with installation



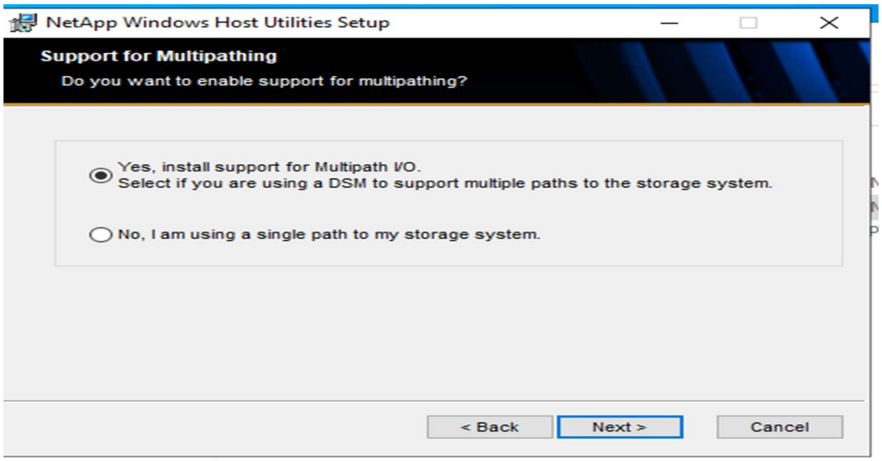
2.Click **OK** to continue.



3. Accept the license agreement and click **Next**.



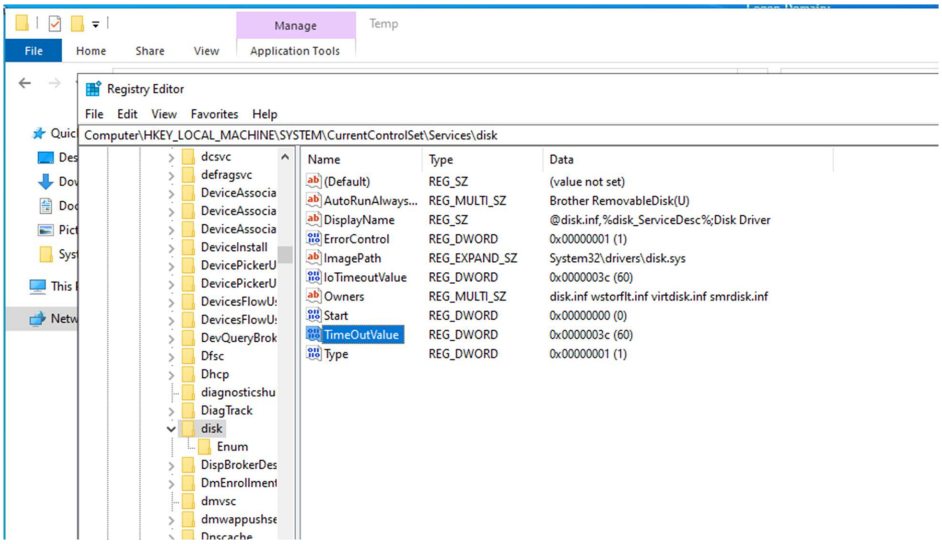
4. When prompted, select: **Yes, install support for Multipath I/O**.



5.After successful installation, a **system reboot is required**.

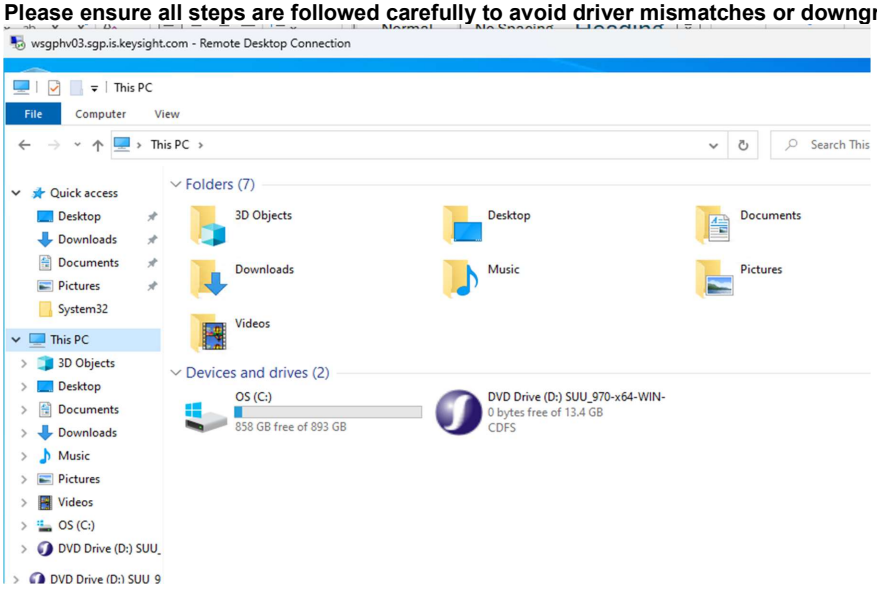
Post-Installation Note:

- The disk timeout value will revert from **120 seconds to 60 seconds** after installation.
- Please correct the disk timeout value back to **120 seconds** as per standard configuration.

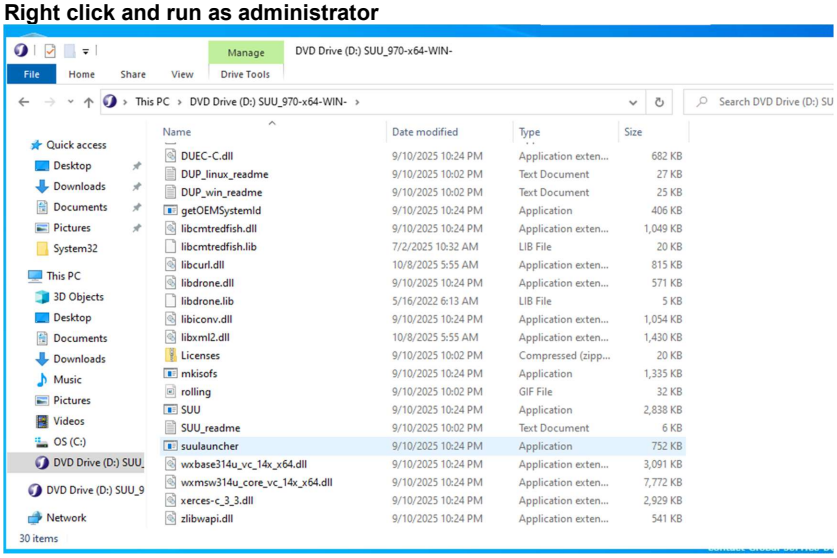


6.3 Drivers upgrade using Dell Server Update Utility

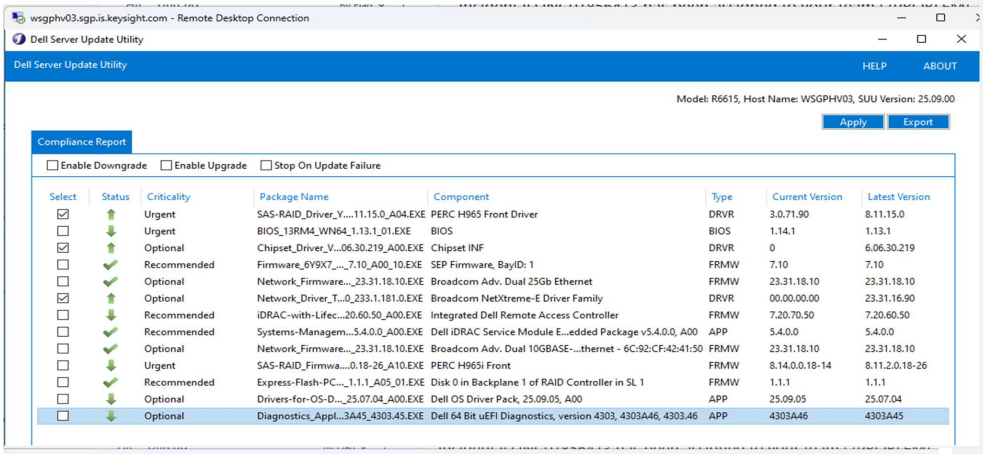
Please follow the steps below to upgrade the network drivers:
Application Location: \\wsgphv02.sgp.is.keysight.com\c$\Temp
Installation Process
Please ensure all steps are followed carefully to avoid driver mismatches or downgrades.

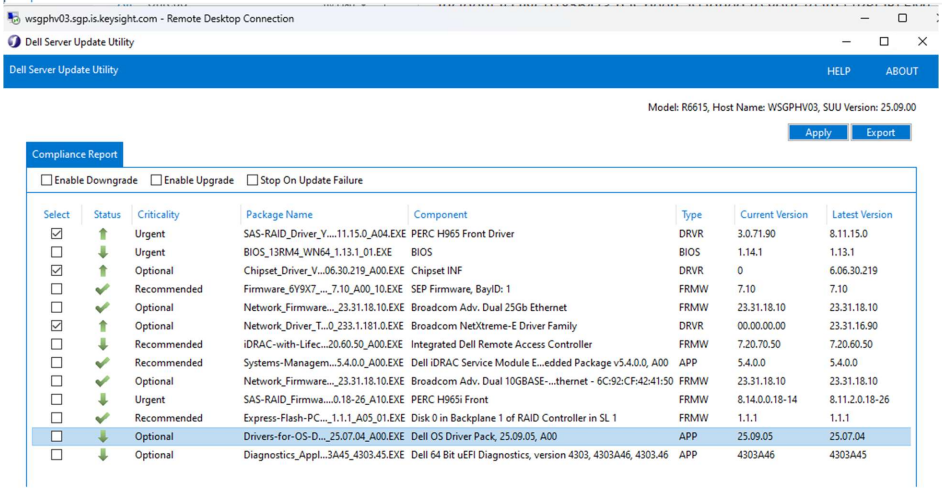


2. **Open the mounted drive and locate SSU Launcher.**
Right click and run as administrator



3. **When the prompt opens, review carefully:**
   - **It will display current driver versions and latest available versions.**
   - **Uncheck all drivers except those with a higher version than the current one.**
   - **Do NOT downgrade any drivers.**

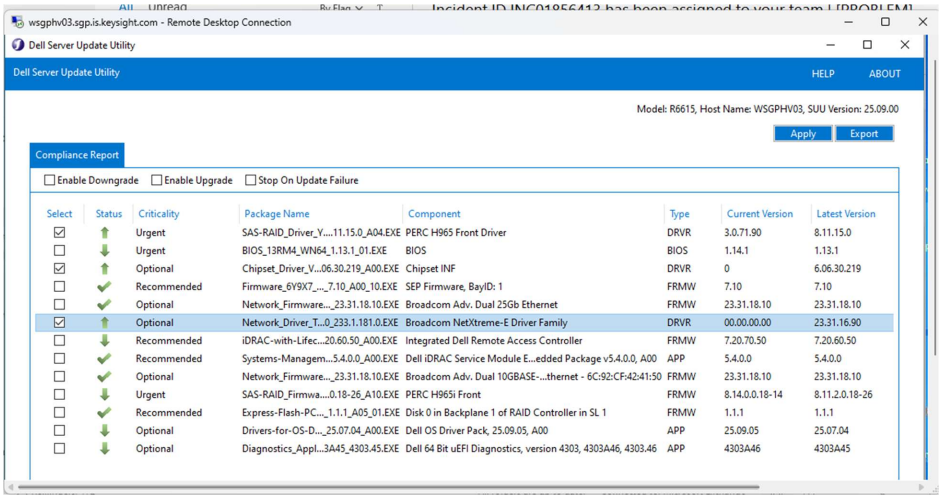




4. Uncheck the following options: if they are checked

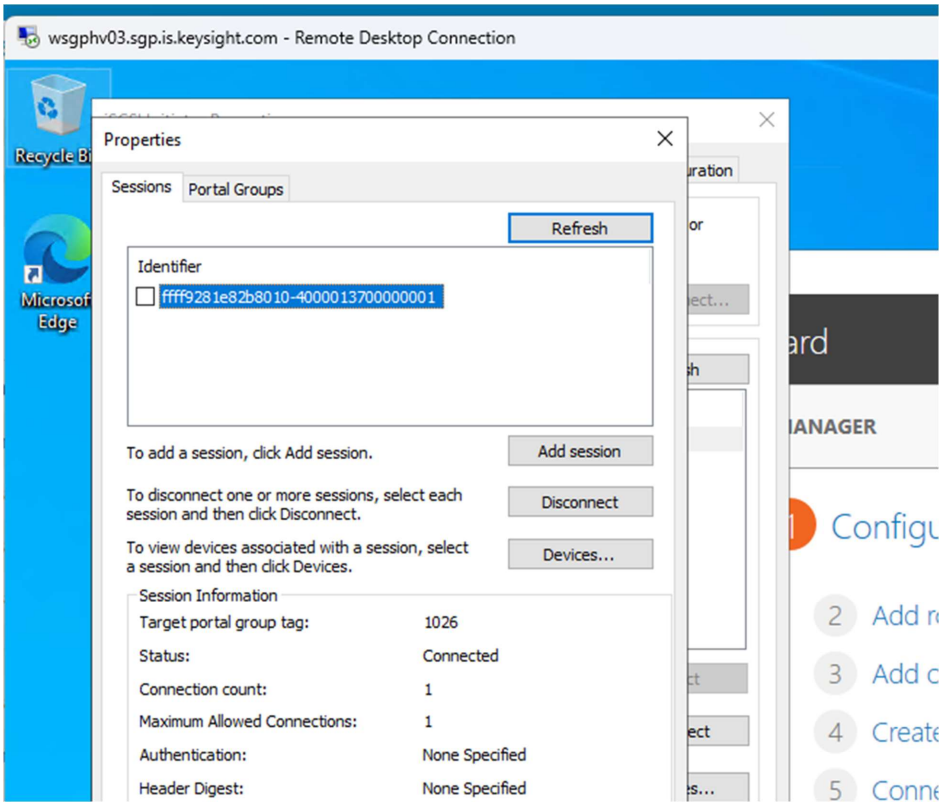
- **Enable downgrade and upgrade**
- **Stop on update failure**

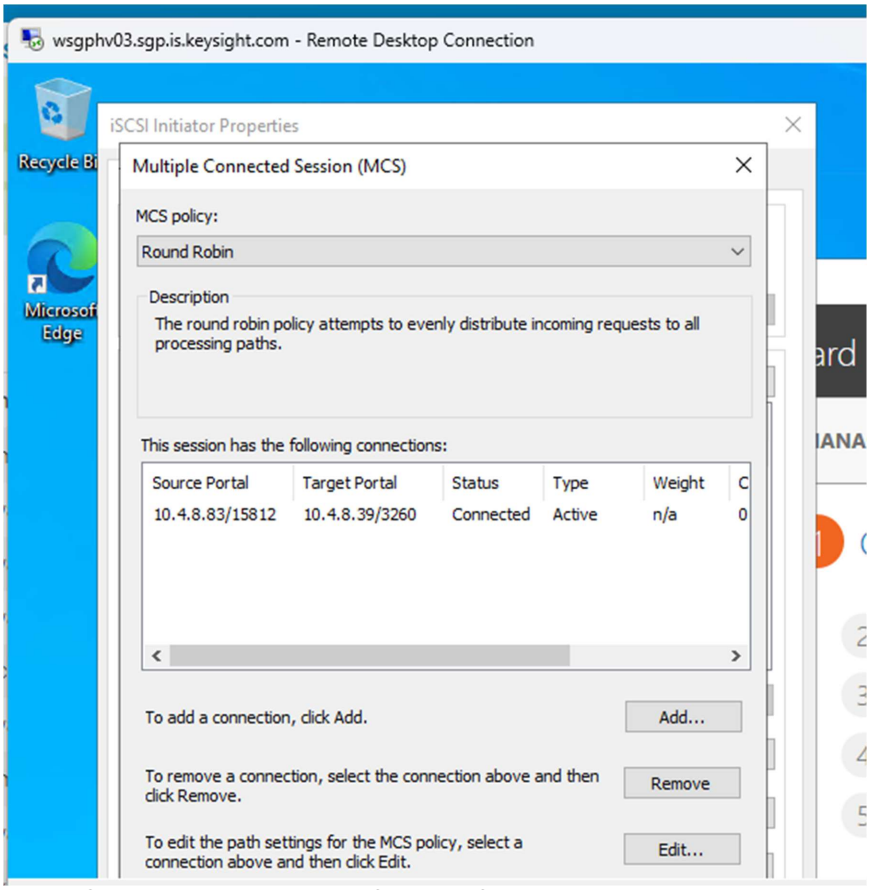
And select only the drivers that need upgrading and click Apply.



Post-Upgrade Checks

1. **Some iSCSI sessions may be removed automatically. Verify and re-add any missing sessions.**


wsgphv03.sgp.is.keysight.com - Remote Desktop Connection
Recycle Bi
Microsof
Edge
Properties
Sessions
Portal Groups
Refresh
Identifier
ffff9281e82b8010-4000013700000001
To add a session, click Add session.
Add session
To disconnect one or more sessions, select each session and then click Disconnect.
Disconnect
To view devices associated with a session, select a session and then click Devices.
Devices...
Session Information
Target portal group tag: 1026
Status: Connected
Connection count: 1
Maximum Allowed Connections: 1
Authentication: None Specified
Header Digest: None Specified
ANAGER
Configu
2 Add r
3 Add c
4 Create
5 Conne



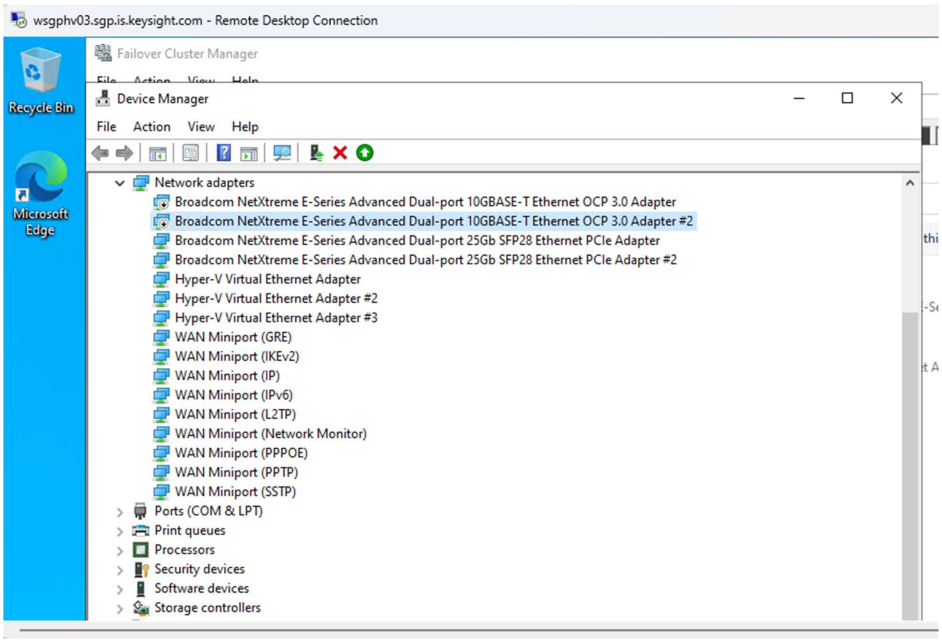
2. **Compare driver versions before and after the upgrade using:**

PowerShell cmd to check: Get-NetAdapter | Select Name, InterfaceDescription, DriverVersion

```
PS C:\Users\laxmcholadm> Get-NetAdapter | Select Name, InterfaceDescription, DriverVersion

Name                    InterfaceDescription                                                    DriverVersion
----                    --------------------                                                    -------------
Integrated NIC 1 Port 2-1 Broadcom NetXtreme E-Series Advanced Dual-port 10GBASE-T Ethernet OCP 3.0 Adapter #2 233.0.148.0
Integrated NIC 1 Port 1-1 Broadcom NetXtreme E-Series Advanced Dual-port 10GBASE-T Ethernet OCP 3.0 Adapter    233.0.148.0
vEthernet (LiveMigration) Hyper-V Virtual Ethernet Adapter #2                                   10.0.20348.1
SLOT 1 Port 2           Broadcom NetXtreme E-Series Advanced Dual-port 25Gb SFP28 Ethernet PCIe Adapter #2   233.0.148.0
SLOT 1 Port 1           Broadcom NetXtreme E-Series Advanced Dual-port 25Gb SFP28 Ethernet PCIe Adapter      233.0.148.0
vEthernet (LS-SGP-S01)  Hyper-V Virtual Ethernet Adapter                                        10.0.20348.1
vEthernet (Storage)     Hyper-V Virtual Ethernet Adapter #3                                     10.0.20348.1
```

3. **Check Device Manager for any unknown devices .**



4.Disable unused network adapters again: After the driver upgrade, previously disabled NICs may be re-enabled automatically. Please review and disable any NICs that are not in use.